Financial Analyst Agent — Engineering Notes

Living notes: architecture decisions, cost/latency experiments, and fixes.

Task

Build a financial analyst agent able to:

1. analyze a company's quarterly report and reliably extract key financial details (income, costs, etc.);
2. for any industry, return the top N public companies by market size;
3. for any industry, explain how AI can cause disruption and name the most common AI use cases.

Architecture & tool selection reasoning

One primary source per task, chosen from evidence rather than convenience.

> ### Task 1 — Quarterly report analysis → SEC EDGAR
>
> A no-document prompt run in Claude (Sonnet-High) was compared against a run with an attached 10-Q. Results were similar; asking the no-document run to cite its resources revealed it referenced Apple's actual Form 10-Q, fiscal Q3 2026 (quarter ended June 27, 2026), from the SEC EDGAR site. This confirmed SEC EDGAR as the correct primary source and justified including an EDGAR MCP server in the tool set.

Task 2 — Top-N companies → Yahoo Finance (yfinance), fallback Alpha Vantage

FMP MCP's free tier lacked the required tool access, so the stack switched to a community yfinance-based MCP server.

Noted risk: Yahoo does not publish or support a public finance API, so yfinance relies on undocumented endpoints that can change or get rate-limited without notice. Alpha Vantage was added as fallback specifically because it is the only official, vendor-maintained financial MCP server currently available — a more dependable backup than another community-built option.

Task 3 — AI disruption / use-case analysis → Tavily

A qualitative, open-web research question — routed to Tavily search/research tools rather than structured financial-data tools.

Agent architecture

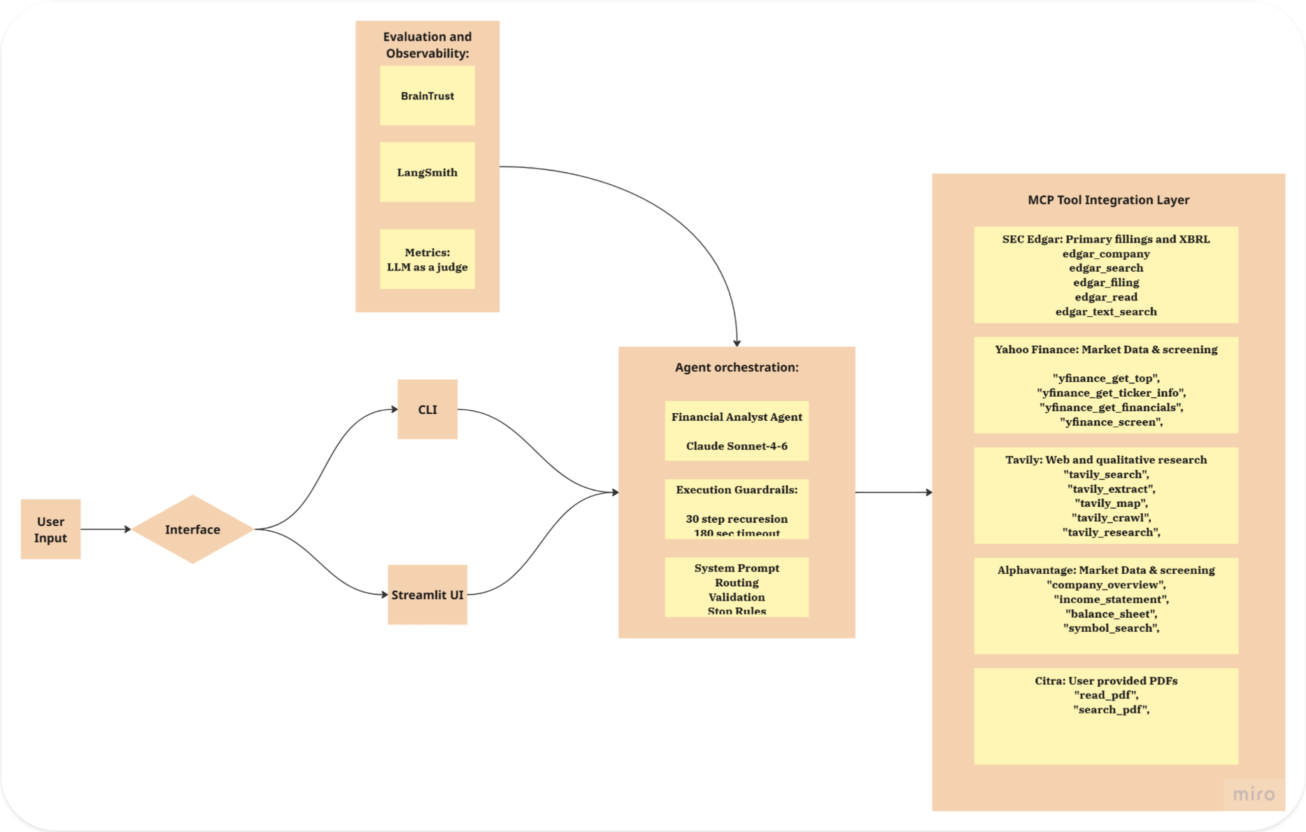

Figure 1. System map. User input arrives through CLI or Streamlit UI and reaches agent orchestration — the Financial Analyst Agent on Claude Sonnet-4-6, with execution guardrails (30-step recursion, 180-second timeout) and a system prompt covering routing, validation and stop rules. Evaluation and observability (BrainTrust, LangSmith, LLM-as-a-judge metrics) wrap the orchestration layer; the MCP tool integration layer exposes SEC EDGAR, Yahoo Finance, Tavily, Alpha Vantage and a user-PDF reader.

Evaluation dataset

A 60-sample (Q&A) evaluation set was built to test the three core capabilities this agent must perform reliably: extracting specific figures from company filings, ranking public companies by market size within an industry, and explaining AI-driven disruption trends within an industry — plus, critically, chaining these together in compound, multi-step prompts (e.g., rank the top N companies in an industry, then extract a metric for each).

Compound prompts are the harder and more differentiating case for this agent, since they require the source-routing, stop-rule and sanity-check logic described elsewhere in this document to fire correctly across multiple sequential tool calls rather than once. The dataset's reference answers are the ground truth the LLM-judge (Observation #11) grades agent output against.

SAMPLE SCHEMA

- **id** — unique integer, 1–60.
- **category** — one of six values (see below).
- **question** — the literal prompt given to the agent.
- **answer** — the reference/ground-truth answer.
- **source** — the primary filing, press release or live data source the answer was verified against, with a URL.
- **difficulty** — easy / medium / hard.
- **grading_type** — *factual* for the large majority, or *behavioral* for two samples graded on agent conduct rather than a matchable answer.

Ranking samples and the compound samples that reuse a ranking additionally carry a **grading_note** field spelling out grading tolerance; edge-case samples carry an **edge_case_type** field naming the specific failure mode targeted.

Category breakdown

CATEGORY	COUNT	EASY	MEDIUM	HARD
single_metric_extraction	12	4	5	3
multi_metric_extraction	11	0	8	3
company_ranking	8	4	4	0
ai_disruption	8	4	4	0
compound_multi_tool	13	0	9	4
edge_case	8	0	4	4
Total	**60**	**12**	**34**	**14**

Edge-case coverage

One sample per failure mode.

1. **No public filings.** A company (Cargill) with no SEC filings at all, testing whether the agent declines rather than substituting a third-party revenue estimate as if it were filing-sourced.
2. **Future unreported quarter.** A fiscal quarter (Nvidia Q2 FY2027) not yet reported as of the dataset's construction date, testing whether the agent declines and cites the actual reporting date rather than fabricating a result or presenting guidance as an actual.
3. **Post-cutoff answerable.** The mirror-image case: a real, already-occurred event (SpaceX's June 2026 IPO) that postdates a typical training cutoff but is fully searchable, testing for the opposite failure mode — a lazy refusal or a confidently-stated stale fact ("SpaceX is private") instead of searching.
4. **Ticker name collision.** Two distinct public companies sharing a name (The Coca-Cola Company vs. Coca-Cola Consolidated), testing whether the agent disambiguates rather than silently picking one or blending the two.
5. **Unanswerable subpart.** A compound question where one company doesn't disclose the requested metric in the requested form, testing whether the agent flags the gap rather than fabricating or quietly substituting a different metric.

6. **Adversarial vague.** A question with no company, period or metric specified, testing whether the agent asks a clarifying question rather than guessing.
7. **Nonstandard industry boundary.** An industry ("medical devices") whose membership genuinely differs across published rankings depending on methodology, testing whether the agent states the ambiguity instead of presenting one aggregator's list as unambiguous fact.
8. **GAAP/non-GAAP embedded in edge framing.** A false-premise question asserting a GAAP figure beat a guidance range actually issued on a non-GAAP basis, testing whether the agent corrects the premise instead of confirming it.

Current results

Latest experiment run over the full 60-sample set, grouped by `metadata.category`.

ANSWERED	LLM JUDGE	DURATION
98.33%	**84.91%**	**3,724.67s**
average	average	sum, all 60 samples

CATEGORY	SAMPLES	% ANSWERED	% LLM_JUDGE	DURATION (SUM)
ai_disruption	8	100%	100%	589.7s
company_ranking	8	100%	78.13%	370.33s
compound_multi_tool	13	100%	85.42%	1,036.14s
edge_case	8	100%	68.75%	274.84s
multi_metric_extraction	11	90.91%	90.91%	995.15s
single_metric_extraction	12	100%	84.09%	458.5s

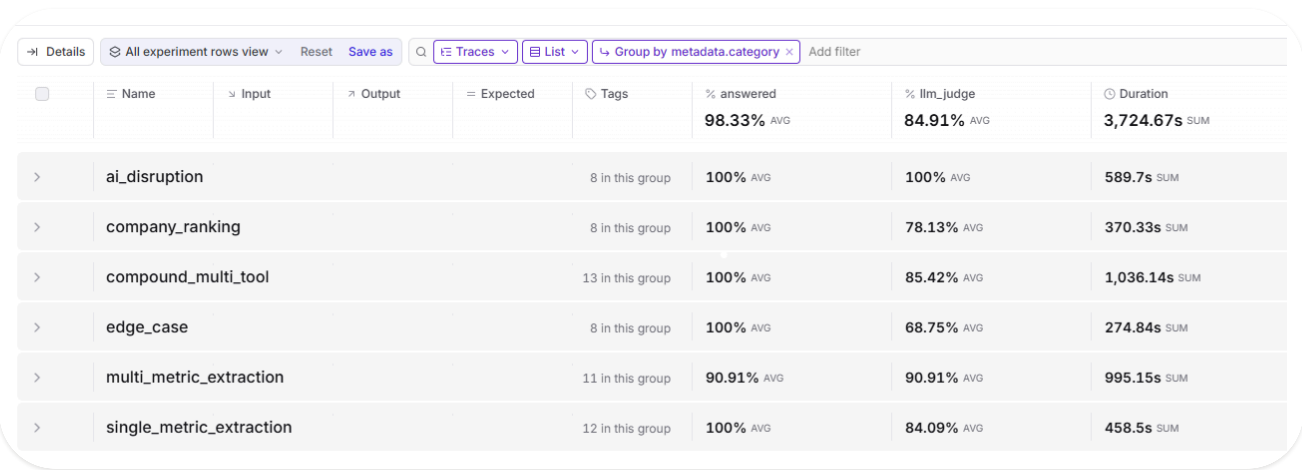

Figure 2. Experiment rows grouped by category in BrainTrust, with the run-level averages in the column headers.

Reading across the run: **ai_disruption** is fully solved (100% judge score), while the two weakest categories are **edge_case** (68.75%) and **company_ranking** (78.13%) — both categories whose grading depends on conduct and on live, freshly-changing data rather than on a fixed filed figure. **multi_metric_extraction** is the only category not fully answered (90.91%, i.e. one sample), and it is also the second most expensive in wall-clock time (995.15s across 11 samples) after **compound_multi_tool** (1,036.14s across 13).

Observations and fixes

Each entry records the symptom observed in traces, the root cause found, and the change made.

1 Redundant full-statement pulls cost ≈ $0.80 per question

Question. "Where did Ford rank among U.S. automakers by revenue in 2024, and what were its Q2 2026 GAAP revenue and net income/loss?" — cost ≈ $0.80 for a single question.

Root cause. Two parallel `edgar_company` calls (GM, Tesla) each returned full income statement, balance sheet and cash flow, rendered as box-drawing ASCII tables with every XBRL line item included — thousands of tokens of mostly irrelevant data for a question that only needed one revenue figure per company. Because this content is freshly generated tool output, prompt caching could not help on the turn it was produced; and because it then lives in conversation history, it gets re-billed at full price on every subsequent turn too.

Fix. Do not let a failed or ambiguous `edgar_trends` call escalate automatically to the much larger `edgar_company(include=["financials"])` response.

2 Cache breakpoint was static, so the cached prefix never grew

Symptom. Caching wasn't done properly. In BrainTrust the cached prompt stayed the same size and did not increase with each tool call — the read-cache portion of each span stayed flat while total tokens climbed, so cache hit fell from 97.3% on the first call to 21.4% by the last.

Figure 3. Before the fix. The hatched cached prefix is the same width on every span (about 13K tokens) while total tokens grow to 60,786; cache hit degrades 97.3% → 53.9% → 42.3% → 31.2% → 21.4%.

Fix. Add a second, moving `cache_control` breakpoint at the end of the message history on every turn.

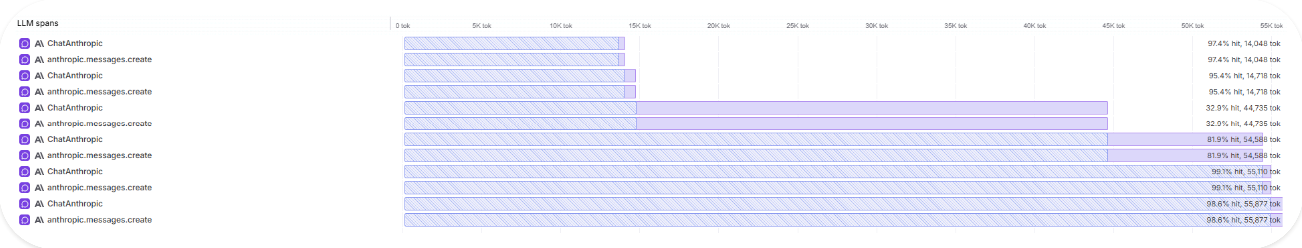

Figure 4. After the fix. The cached prefix now grows with the conversation; after a mid-run dip to 32.9%, cache hit recovers to 81.9%, then 99.1% and 98.6% on spans of 55,110 and 55,877 tokens.

Result. Cache hit tracks the conversation instead of decaying, with a large improvement in estimated cost.

3 Self-verification treated as a quality bar

Symptom. The agent routinely called a second or third source (Yahoo Finance, Tavily) to "confirm" a figure it had already retrieved cleanly from the stated primary source, inflating both tool-call count and token cost. Trace language literally said "I now have confirmed, precise data from both SEC EDGAR and Yahoo Finance."

Root cause. The system prompt said sources are "preferred," not that the agent must stop once a sufficient result is obtained. "Insufficient evidence" was left as a subjective judgment call for the model, and a soft "attempt ONE alternative" budget was ambiguous about what counts as one attempt.

Fix. An explicit, numeric stop rule — maximum 2 tool calls total per requested metric (1 primary + 1 fallback), across the entire resolution of that metric, not per tool. "Sufficient" was defined objectively (a value with a clear period and form-type label) rather than left to the model's discretion.

4 Correct call, wrong number: MSFT FY2026 revenue overstated 2×

Scenario. "What was Microsoft's total revenue for fiscal year 2026 (fiscal year ended June 30, 2026)?"

RETURNED	TRUE REPORTED
$684,000,000,000	**$331,839,000,000**
stated YoY growth 142.8%	≈17.8–18% YoY

Symptom (accuracy). `edgar_trends(concepts=["revenue"], identifier="MSFT", period="annual", periods=2, include_growth=true)` — called with parameters fully correct per the routing matrix — returned the figures above.

Root cause. Most likely duplicate or overlapping XBRL duration-context facts tagged under the same revenue concept and period label (e.g. a full-year fact and a partial-year or restated fact both matching "2026") being summed or mis-selected by `edgar_trends`'s aggregation logic.

Fix. Added a `sanity_check_rule` section defining explicit plausibility bounds — YoY/QoQ growth outside approximately −50% to +100% for an established company is flagged.

5 17.8% vs. 18%: a precision mismatch caused by truncated MD&A

Scenario. The same MSFT FY2026 revenue query, during the Observation #4 fallback path.

Symptom (precision, not correctness). The final answer reported revenue growth as "17.8%" — self-calculated from the two raw revenue figures — rather than Microsoft's own stated "18%", the figure a reference answer built from company language would use. Both are consistent (17.79% rounds to 18%) and neither is factually wrong, but the mismatch reads as a discrepancy against precision-sensitive reference answers.

Root cause. Inspecting the raw `edgar_read(sections=["financials","mda"])` output directly: the returned MD&A text includes the "Overview" and bulleted "Highlights" subsections (segment-level growth only — Cloud +27%, Azure +41%, etc.) but is truncated mid-document, ending with "... (truncated)" before reaching "Results of Operations", where total-company revenue growth is conventionally stated in prose ("Revenue increased $X billion or 18%"). The agent did not ignore an available stated figure; the figure was never present in the text it had access to.

Fix. None yet at the tool level.

6 Scores dropped because the ground truth had gone stale

Question. "List the top 5 U.S. banks by market capitalization as of today, with each bank's market cap."

Symptom. Although the agent had previously produced precise results for this category of question, it stopped generating correct answers and the LLM judge gave lower scores for them.

Fix. Reviewing the generated answers showed the agent was still providing correct answers; because of the live nature of these questions, the ground-truth answer had gone out of date. After fixing the ground truth, the agent scored correct results again.